1 Program

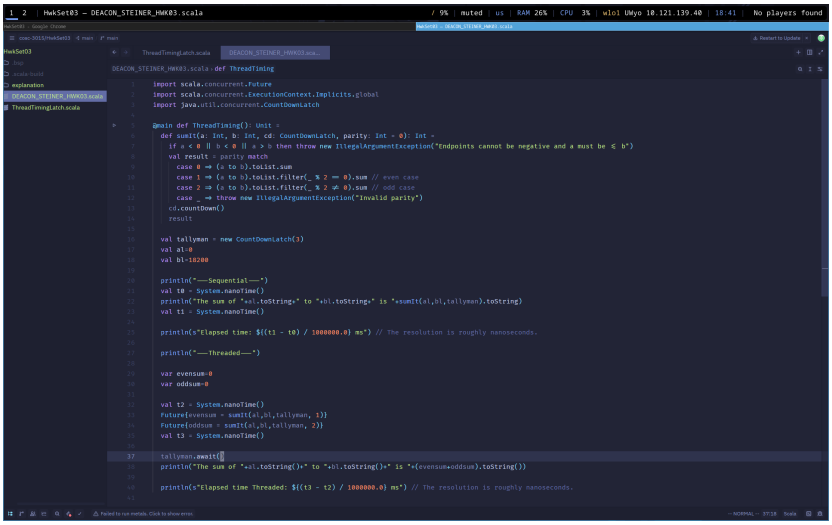


Figure 1: Assignment Solution

This program was relatively easy to adapt. I used pattern matching along with filter to make `sumIt` a very functional solution to this problem. I use a parity parameter to determine whether the function should sum the whole list (parity = 0) the evens (parity = 1) or the odds (parity = 2). I do the split using `filter` which is a very functional way to build a sublist that follows the needed criteria.

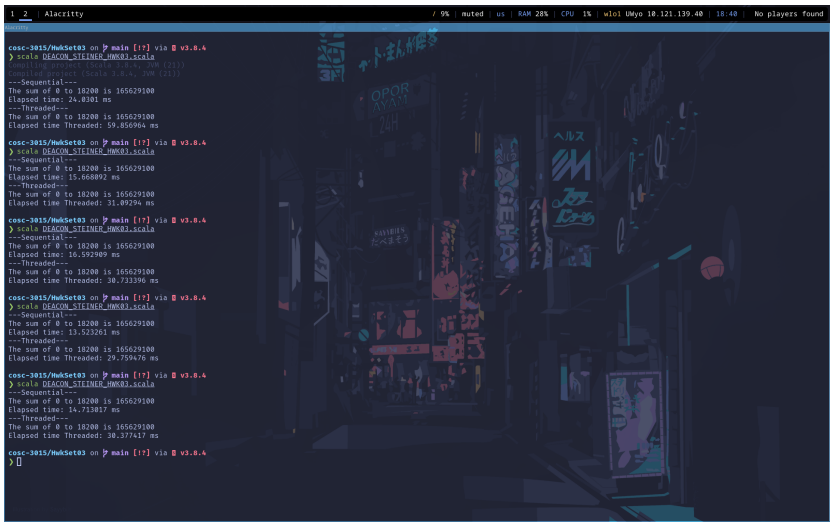


Figure 2: Program output

This output shows that the threaded option is actually slower (sometimes by more than 20 ms) but this is likely explained by the fact that `sum` is fairly efficient for small ranges and the overhead of scheduling and synchronizing concurrent tasks can outweigh the benefits of parallel execution.